

F I N T E C H • B U S I N E S S A N A L Y S I S C A S E
S T U D Y

Merchant Onboarding, Payment Processing & Risk Management

Designing a Secure, Scalable & Data-Driven Merchant Acquiring Platform

P R E P A R E D B Y

Vaibhav Yadav

Business Analyst | Case Study Portfolio

PostgreSQL • SQL • Tableau • Figma • REST APIs

Executive Summary

BUSINESS CONTEXT

Traditional merchant acquiring platforms rely on fragmented onboarding, manual KYC verification, delayed fraud reviews, and slow settlements—resulting in higher operational costs and delayed merchant activation.

PROPOSED SOLUTION

An integrated platform that automates merchant onboarding, KYC verification, payment processing, fraud monitoring, and settlement management through a centralized workflow.

KEY BUSINESS OBJECTIVES

► Onboarding

Reduce merchant onboarding turnaround time

► Compliance

Automate KYC verification & compliance workflows

► Risk

Strengthen fraud detection & transaction monitoring

► Settlement

Improve settlement efficiency & operational visibility

► Insights

Enable data-driven decisions via executive dashboards

► Scale

Support scalable, secure, cloud-ready architecture

PROJECT SCOPE

BRD • Functional/Non-Functional Requirements • AS-IS / TO-BE • Architecture & ERD • APIs • Agile Stories & UAT • SQL • Tableau

Business Context & Problem Statement

INDUSTRY CONTEXT

Rapid growth in digital payments has increased demand for secure, efficient merchant acquiring. Businesses now expect faster onboarding, seamless payments, proactive fraud monitoring, and timely settlements — all while maintaining regulatory compliance.

CURRENT CHALLENGES



Manual Onboarding

Delays merchant activation



Lengthy KYC

Impacts customer experience



Limited Fraud Visibility

Increases financial risk



Settlement Delays

Hurts merchant cash flow



Fragmented Processes

Reduces operational efficiency

BUSINESS IMPACT

Challenge	Business Impact
Slow onboarding	Delayed merchant activation
Manual KYC	Higher operational effort
Fraud detection delays	Increased financial exposure
Settlement delays	Merchant dissatisfaction
Limited reporting	Poor business decisions
Disconnected systems	Ops inefficiency

Business Objectives & Success Metrics

Design a proposed secure, scalable merchant acquiring platform that automates onboarding & KYC, reduces manual effort, improves payments, strengthens fraud controls, accelerates settlement, and enables operational visibility.

SUCCESS METRICS • KPI SCORECARD

Merchant Onboarding

Merchant Onboarding Turnaround Time

Merchant Activation

Merchant Activation Rate

KYC & Compliance

KYC Approval Rate

Fraud & Risk

Fraud Detection Rate

Settlement

Settlement Processing Time

Settlement Accuracy

Settlement Accuracy

Automation

Operational Automation Rate

Payments

Transaction Success Rate

Dashboard Availability: Static Reports → Real-Time Dashboard | Fraud Detection Response: Manual Review → Real-Time

Stakeholder Matrix & Collaboration Model

STAKEHOLDER MATRIX

Stakeholder	Role	Primary Responsibility
Product Manager	Business Sponsor	Product vision & roadmap
Senior Business Analyst	Requirements Lead	Elicitation & documentation
Engineering Lead	Technical Owner	Solution design & APIs
Risk & Compliance	Compliance Owner	KYC policies & fraud controls
Merchant Operations	Ops Team	Onboarding & settlements
Finance Team	Financial Ops	Reconciliation & reporting
QA Team	Quality Assurance	Testing, validation & UAT
Third-Party Providers	Integration Partner	KYC, payments, notifications

COLLABORATION WORKFLOW



AS-IS Business Process

Illustrating today's merchant onboarding & payment process — manual steps, delays, operational inefficiencies.



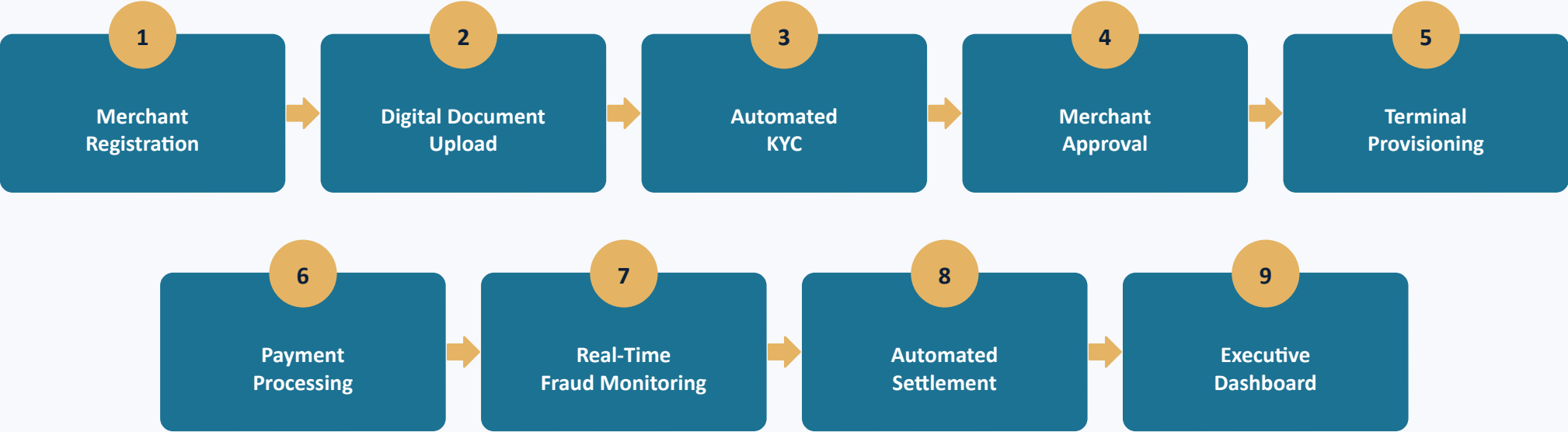
Manual / high-friction step

BUSINESS CHALLENGES & IMPACT

Challenge	Business Impact
Manual KYC Verification	Longer onboarding cycle
Multiple Manual Reviews	Increased operational effort
Delayed Merchant Activation	Slower revenue realization
Reactive Fraud Detection	Higher financial exposure
Manual Settlement	Delayed merchant payouts
Limited Reporting	Poor operational visibility

TO-BE Business Process

A streamlined, automated lifecycle — digital onboarding, real-time fraud monitoring, faster settlement, integrated reporting.

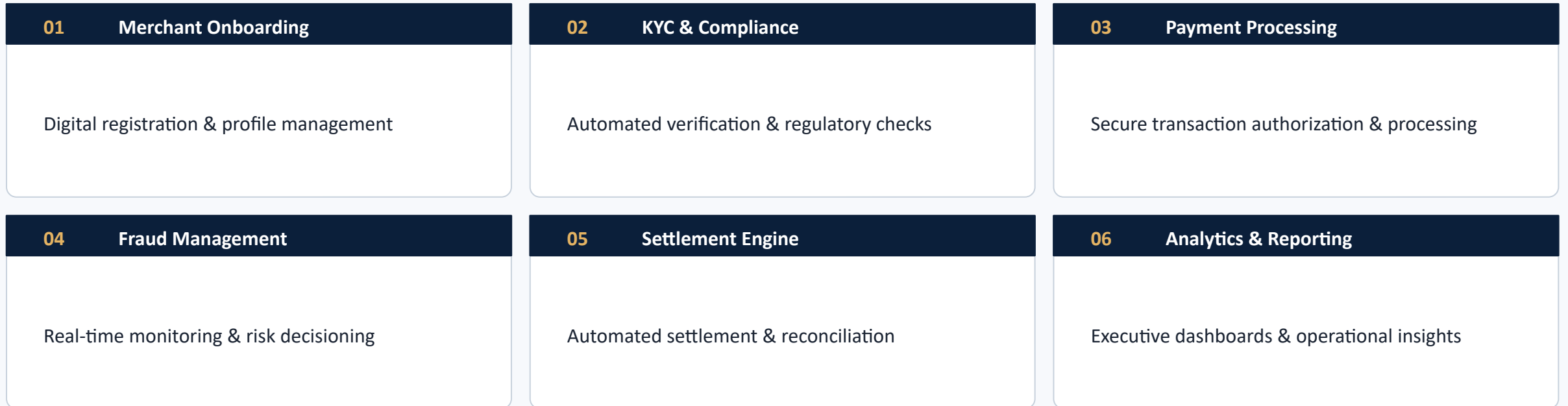


KEY IMPROVEMENTS

Current State	Future State
Manual KYC	Automated KYC Verification
Multiple Manual Reviews	Rule-Based Approval Workflow
Reactive Fraud Checks	Real-Time Fraud Monitoring
Manual Settlement	Automated Settlement
Limited Reporting	Executive Analytics Dashboard

Proposed Solution Overview

A high-level view of the FinTech platform — how business capabilities combine to support the full merchant lifecycle.

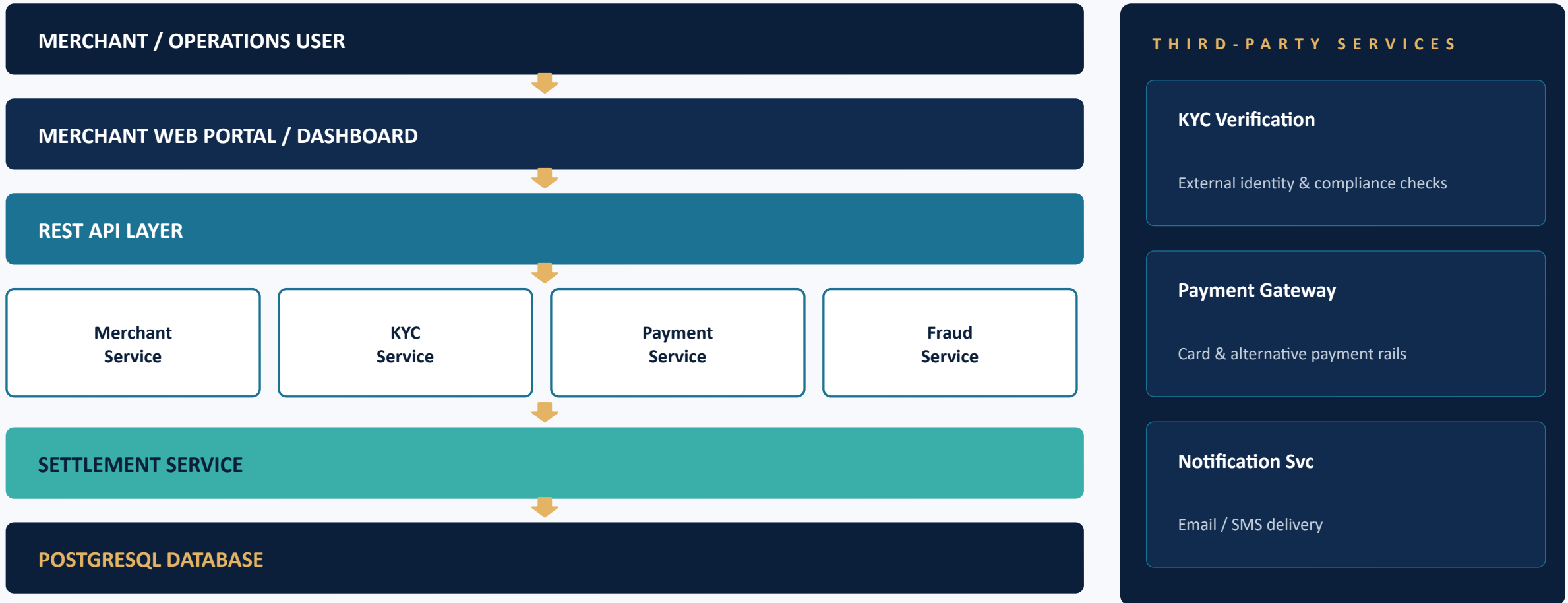


EXPECTED BUSINESS VALUE



High-Level System Architecture

How users, business applications, external services, and the database interact across the merchant lifecycle.

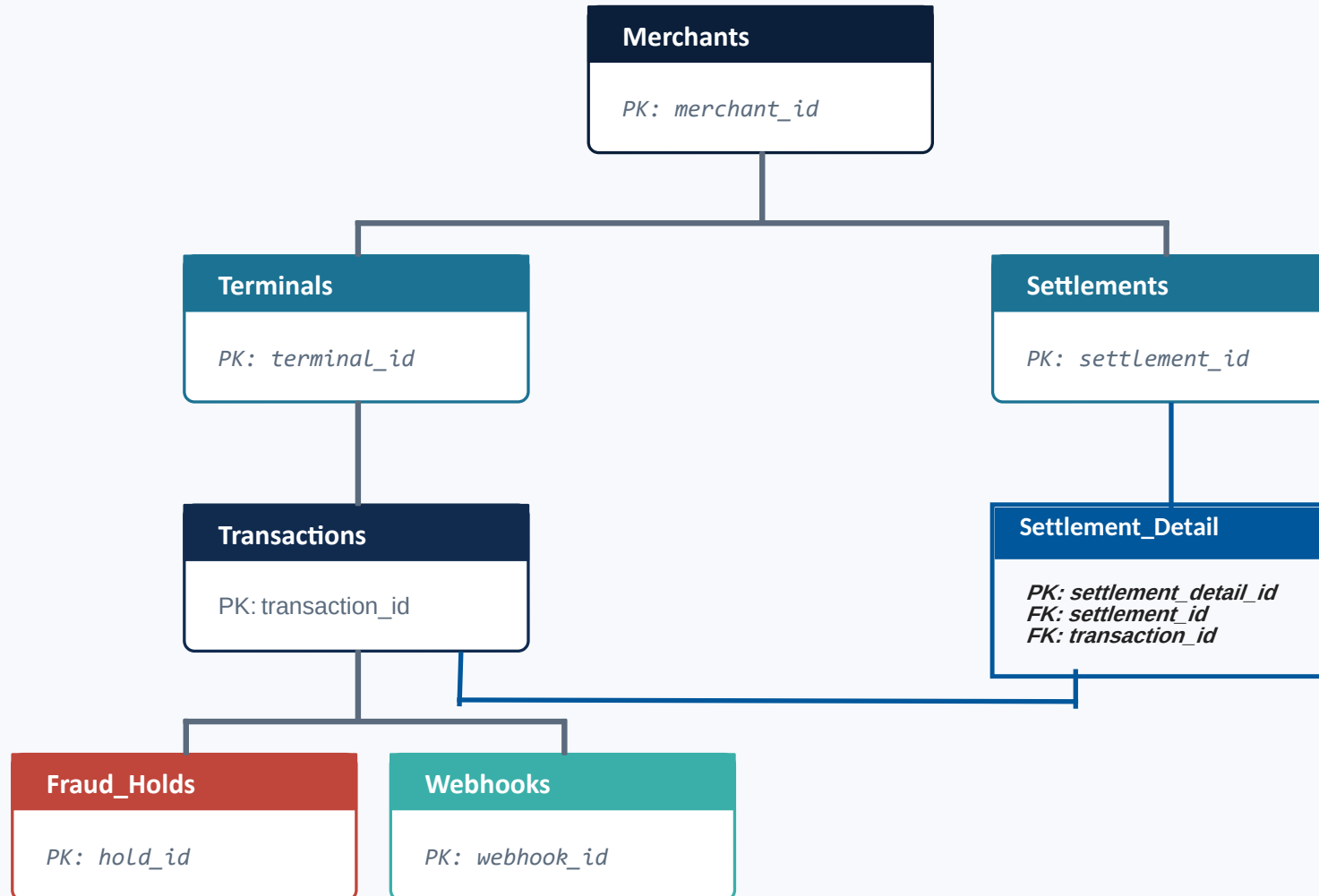


TECH STACK

Frontend: Web Portal • Backend: REST APIs • Database: PostgreSQL • Analytics: SQL + Tableau

Entity Relationship Diagram (ERD)

The logical data model supporting onboarding, payments, fraud management, and settlements.

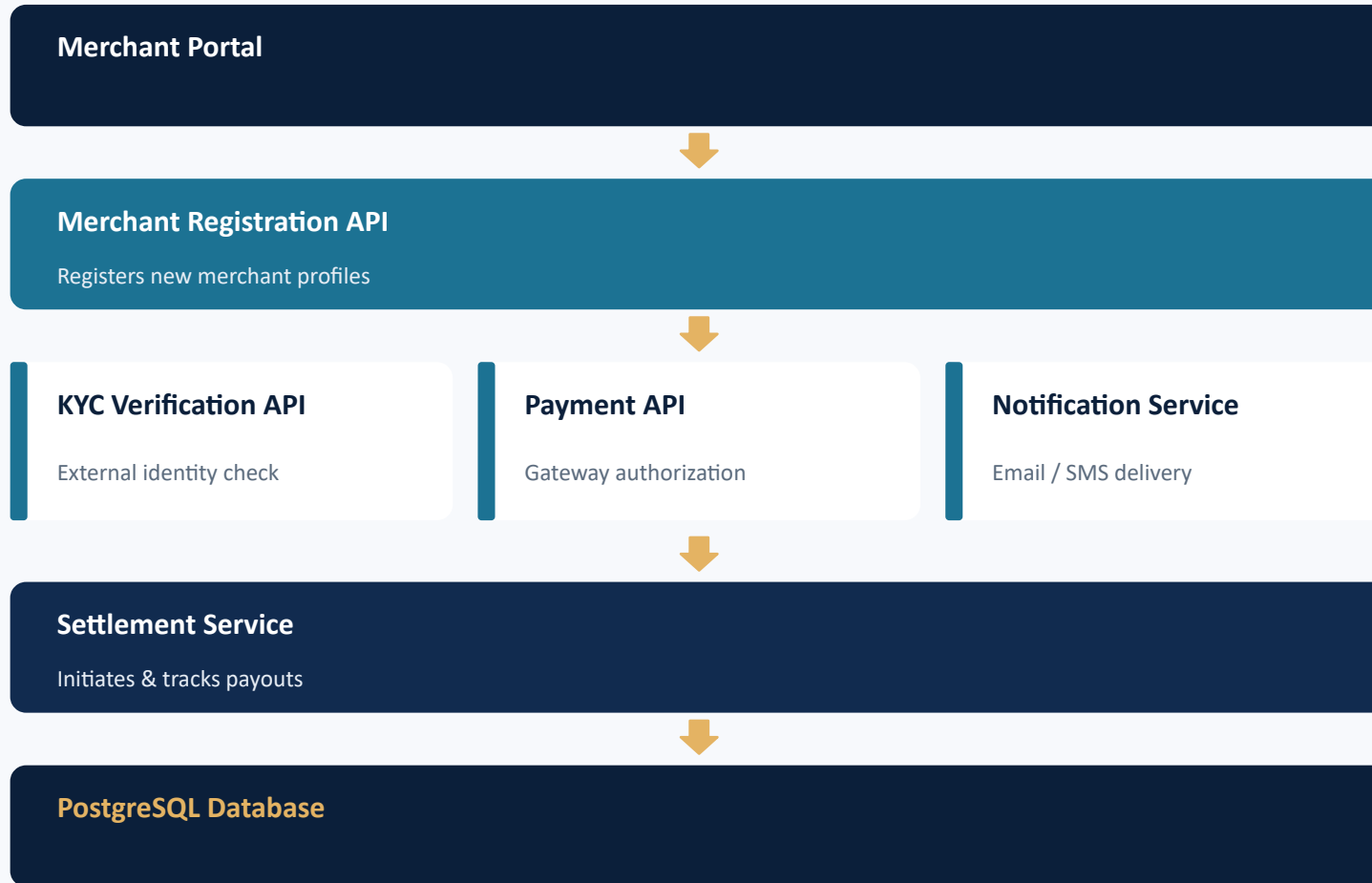


BUSINESS ENTITY SUMMARY

Entity	Business Purpose
Merchants	Profile & KYC status
Terminals	Merchant-to-terminal mapping
Transactions	Payment activity
Fraud Holds	High-risk transactions
Settlements	Merchant payouts
Webhooks	Event notifications & delivery

API Integration Overview

How the platform exchanges information with internal services and external providers.



API LIFECYCLE

- 1 Merchant Registration
- 2 KYC Verification
- 3 Payment Processing
- 4 Settlement Processing

BUSINESS VALUE

Faster Integration

Standardized APIs reduce manual processing

Reliable Payments

Secure gateway integration

End-to-End Visibility

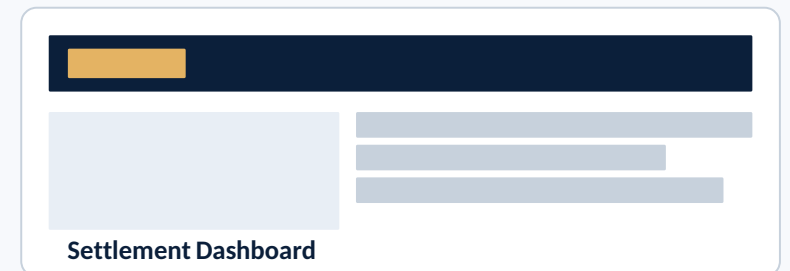
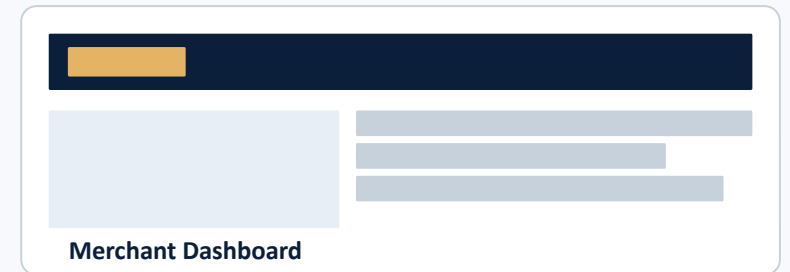
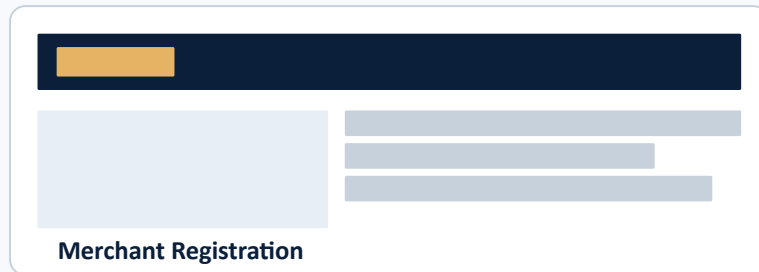
Consistent operational tracking

Product Wireframes & User Journey

End-to-end merchant journey — from onboarding to settlement — with low-fidelity wireframes.



WIREFRAME GALLERY



VALUE

Simplified Merchant Experience • Digital KYC • Centralized Payments & Settlements • Real-Time Visibility

Agile Delivery & Requirements Traceability

Translating business requirements into user stories, validating via acceptance criteria, verifying via UAT.

AGILE LIFECYCLE

1

Business Requirements
BRD & FR / NFR

2

User Stories
Sprint backlog input

3

Gherkin Acceptance Criteria
Given / When / Then

4

Sprint Development
Iterative delivery

5

UAT Testing
Business validation

6

Business Sign-off
Solution readiness

SPRINT SNAPSHOT

Sprint 1

Merchant Onboarding & KYC

Sprint 2

Payment Processing & Fraud

Sprint 3

Settlement & Reporting

REQUIREMENTS TRACEABILITY MATRIX

BR	FR	US	UAT
BR-01	FR-01	US-01	UAT-01
BR-02	FR-02	US-02	UAT-02
BR-03	FR-03	US-03	UAT-03

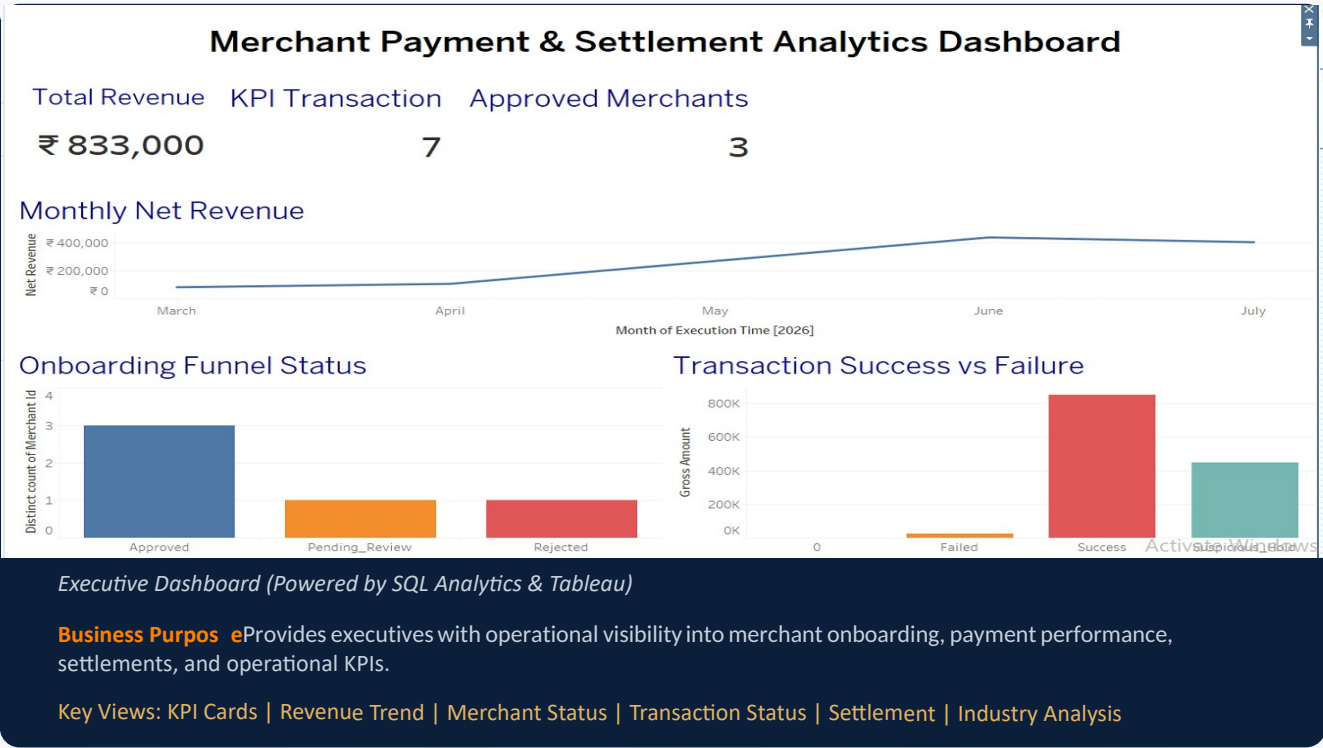
Business Analytics & Executive Dashboard

SQL analytics and Tableau dashboards supporting operational and financial decisioning across the merchant lifecycle.

SQL BUSINESS ANALYTICS • 8 QUERIES

Business Report	Business Insight
Monthly Revenue Trend	Monitor revenue growth
Merchant Onboarding Status	Track KYC pipeline
Transaction Success Rate	Measure payment performance
Fraud Hold Analysis	Monitor high-risk txns
Settlement Summary	Track payout status
Merchant Revenue Ranking	Top-performing merchants
Industry Volume Analysis	Compare business segments
Processing Fee Analysis	Evaluate operational costs

EXECUTIVE DASHBOARD



BUSINESS INSIGHTS DELIVERED

Revenue Performance

Volume & merchant revenue

Risk Visibility

Fraud trends & suspicious txns

Operational Efficiency

Onboarding & settlement

Executive Decisions

Real-time KPI monitoring



Business Impact & Project Outcomes

Expected business improvements from the proposed merchant acquiring platform.

BEFORE → AFTER

Business Area	Current State	Future State
Merchant Onboarding	Manual	Automated
KYC Verification	Manual Review	Automated Validation
Fraud Monitoring	Reactive	Real-Time Detection
Settlement Processing	Manual/Delayed	Automated Settlement
Business Reporting	Static Reports	Interactive Dashboard

SUCCESS METRICS

Faster Merchant Activation

Reduce merchant onboarding turnaround time!

Improved Transaction Reliability

Improve transaction processing success

Faster Settlement Processing

Improve settlement efficiency and processing time

Improved Operational Visibility

Provide centralized dashboards for onboarding, payments, fraud and settlement

Faster Operations • Improved Compliance • Better Operational Efficiency • Data-Driven Decisions

Project Deliverables & Technology Stack

Consolidated view of project artefacts and technologies across the Business Analysis lifecycle.

BUSINESS ANALYSIS

- Business Requirements Document (BRD)
- Functional & Non-Functional Requirements
- Business Rules
- Stakeholder Matrix
- AS-IS & TO-BE Process Models

SOLUTION DESIGN

- System Architecture
- Entity Relationship Diagram (ERD)
- API Contracts
- Low-Fidelity Wireframes
- Agile User Stories
- Gherkin Acceptance Criteria
- UAT Test Scenarios

ANALYTICS

- PostgreSQL Database Design
- SQL Business Analytics (8 Queries)
- Tableau Executive Dashboard
- Repository Assets

TECHNOLOGY STACK

Category	Technology
Database	PostgreSQL
Query Language	SQL
Dashboard	Tableau Public
Wireframing	Figma
Documentation	Microsoft Word
Presentation	Microsoft PowerPoint
Repository	GitHub
API Docs	JSON / REST APIs

REPOSITORY ASSETS

`schema.sql`

`sample_data.sql`

`fintech_queries.sql`

`README.md`

Key Learnings & Future Roadmap

KEY LEARNINGS

01

Stakeholder Collaboration

Aligned Product, Engineering, Ops, Risk & Compliance.

02

Requirements Engineering

Business needs → FRs, rules, user stories, AC.

03

Process Improvement

Redesigned lifecycle by eliminating manual activity.

04

API & System Integration

External KYC, gateway, notifications.

05

Data-Driven Decisions

SQL analytics + dashboards for reporting.

06

Agile Delivery

Sprints, Gherkin, UAT, business sign-off.

FUTURE ROADMAP

NOW

Multi-Currency Settlement

Support international merchant payouts.

NEXT

Predictive Fraud Analytics

ML-based fraud pattern detection.

LATER

Merchant Self-Service Portal

Onboarding & settlement transparency.

VISION

Advanced Executive Analytics

Forecasting & trend analysis.

Questions & Discussion

This case study demonstrates an end-to-end Business Analysis approach for designing a secure, scalable, and data-driven Merchant Acquiring Platform — covering business requirements, solution design, Agile delivery, system architecture, analytics, and executive reporting.

PROJECT SUMMARY

Merchant Onboarding, Payment Processing & Risk Management Platform

Business Analyst Case Study
Prepared by **Vaibhav Yadav**